



Inria, Paris, France

[Date]

Professor Giovanni Neglia PhD Position F/M Distributed Training of Machine Learning Models with Malicious Clients Inria, Paris, France

Dear Professor Neglia,

I am excited to apply for the PhD position on Distributed Training of Machine Learning Models with Malicious Clients at Inria in Paris, France. As a data science engineer with a strong background in machine learning and programming, I believe my skills and experience make me an ideal candidate for this role.

In my previous research project, "Cardiovascular Medical Digital Twin", I had the opportunity to develop innovative algorithms that demonstrate relevance to this PhD position. For instance, I designed a linear-time state space model called *MedicalMambaT* *that enables sub-millisecond bedside inference. This achievement shows time applications, which is crucial in distributed training settings where malicious clients can manipulate*

Furthermore, I have experience with Physics-Informed Operator Networks (PI-DeepONet) that utilize PyTorch Autograd to ensure physical constraints are respected during model training. This expertise will be valuable in defending against advanced privacy attack strategies and developing robust defenses for federated learning systems.

I am particularly drawn to this PhD position because of the focus on privacy and security in federated learning. My previous work has shown me the importance of considering malicious clients when designing machine learning models, and I believe my skills will enable me to make significant contributions to this field.

Thank you for considering my application. I look forward to discussing my qualifications further.

Sincerely,

Ismail Aissaoui